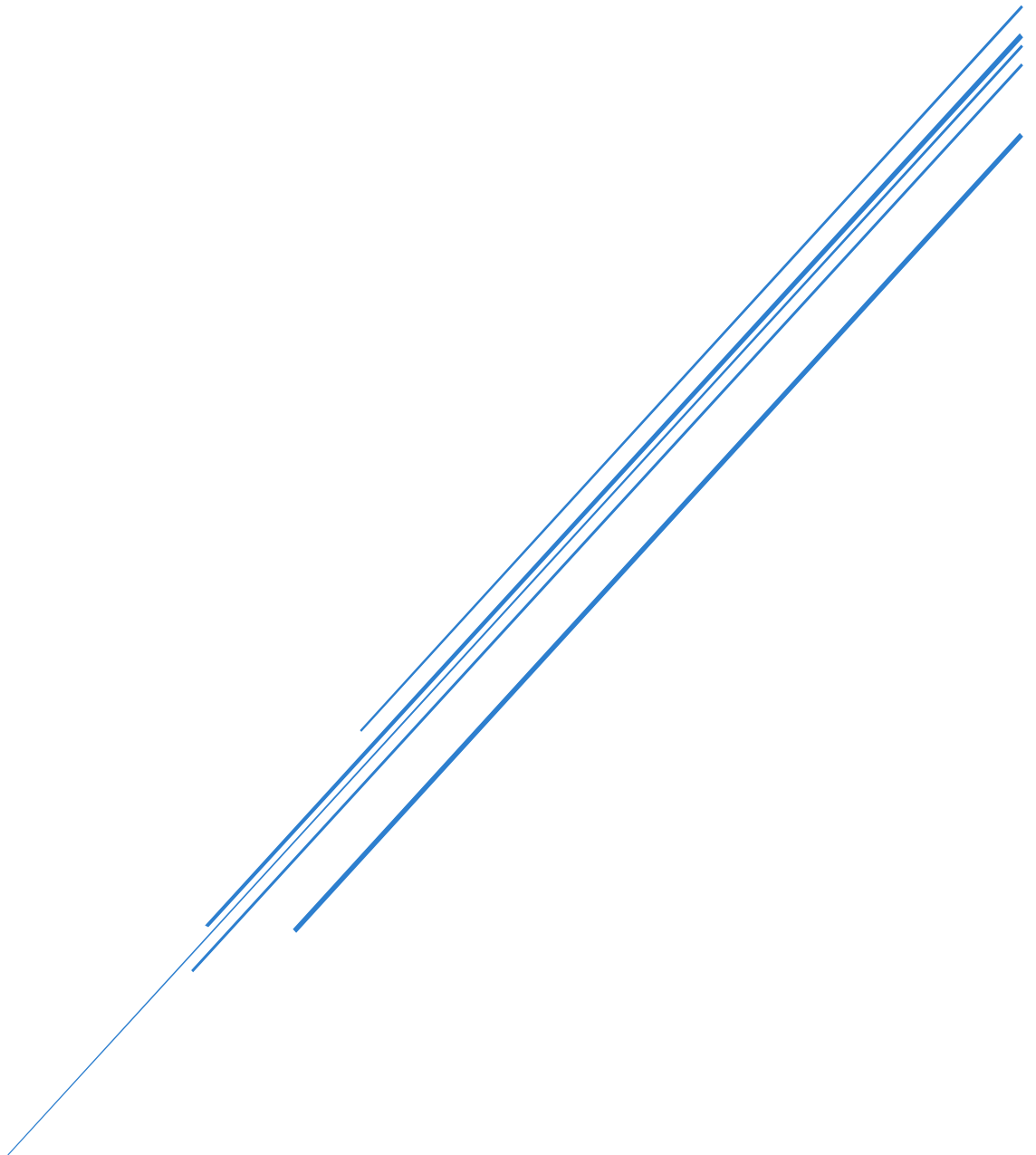


LAB 4 ANSWERS REPORT

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Artificial Intelligence (COMP3742)

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For all these questions the implementation is found in question<x>.py in the Source/ directory.

Question 1

Implement Logistic Classification for binary classification using the Car dataset. Evaluate the model using accuracy and plot the confusion matrix.

Accuracy was 0.8959537572254336.

Confusion matrix doesn't format nicely when pasted into word refer to question1.py output.

Question 2

Implement Logistic Classification for multiclass classification using the Iris dataset. Use cross-validation to evaluate the model's performance.

Cross-Validation Scores: [0.96296296, 1, 1, 0.96296296, 0.88888889]

Mean CV Score: 0.962962962962963

Question 3

Implement Logistic Classification with L1 regularization on the Ionosphere dataset. Evaluate the model using ROC AUC score and plot the ROC curve.

ROC AUC Score: 0.9491304347826087

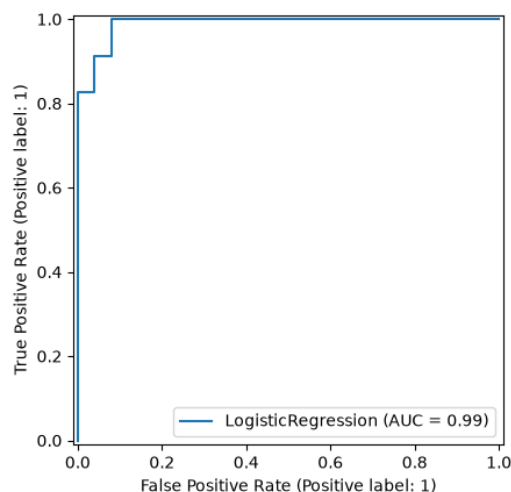


Figure 1: ROC Curve

Question 4

Implement Logistic Classification with class weights on an imbalanced dataset (e.g., the Wine Quality dataset). Evaluate the model using log loss and plot the precision-recall curve.

Log Loss: 0.5389284581213243

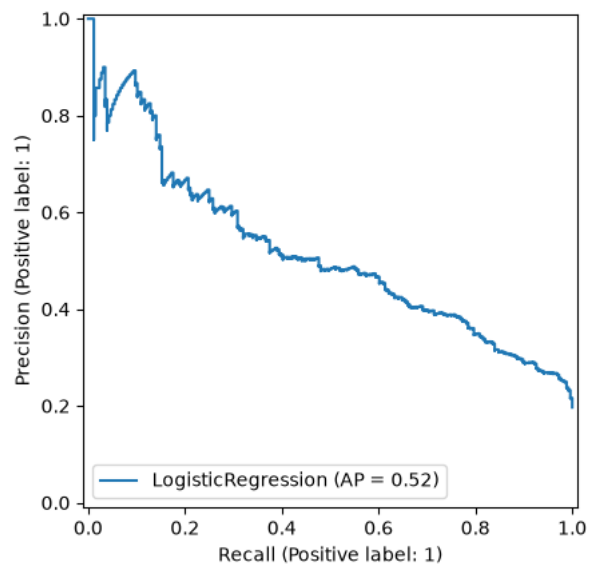


Figure 2: Precision Recall Curve

Question 5

Implement Logistic Classification with different solvers (liblinear, lbfgs, saga) on the MNIST dataset. Compare the performance of each solver using cross-validation and visualize the results.

Best solver was lbfgs with an accuracy of 0.8982, note all solvers were within 0.0001 of each other.

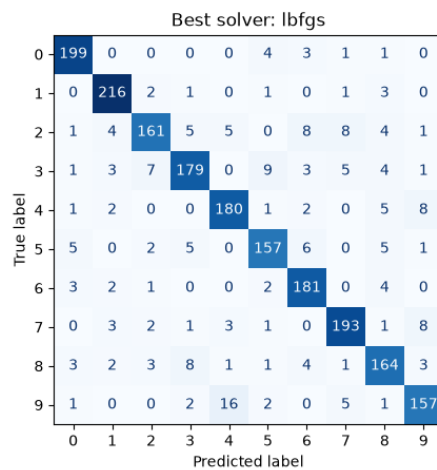


Figure 3: Confusion Matrix